

CE80362 : Circuit Analysis

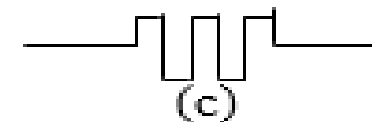
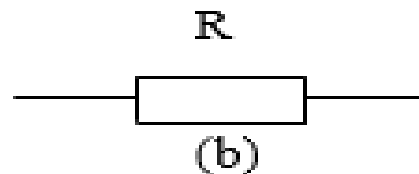
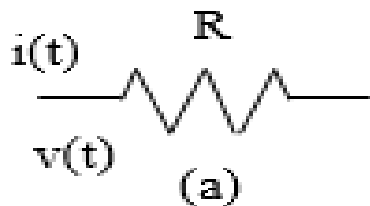
Lecture 1: Passive circuit elements

Definition.

- The circuit elements can be categorized into two types,
- active elements
- passive elements.
- A Passive element is an element which cannot deliver more energy to the circuit than what is supplied to it by the circuit.
- Resistors, capacitors, inductors
- An Active element, delivers more energy to the circuit than what the circuit supplies to it.
- All Power sources (i.e. current or voltage sources) are active elements
- Passive elements do not generate (convert from non-electrical energy) any electricity. They may either consume energy

1. Resistance

- Electrical resistance is the property of a material by virtue of which it opposes the flow of electrons through the material.
- Thus the resistance restricts the flow of electric current through the material.
- The unit resistance being “ohm” (Ω) quantitatively it is obtained from ohm’s law
- The Circuit symbols for Resistance are shown in the figures below.



The Ohm's Law

- This law was first described by Professor Georg Simon Ohm (1791-1867) in a pamphlet.
- It states that “physical condition being unchanged, the current through any conductor is directly proportional to the applied potential difference across it”.
- ie $V \propto I$
- Or $V=IR$, R being the constant of proportionality termed as Resistance
- Therefore $R = \frac{V}{I}$

The Ohm's Law contd'

- When an electric current flows through any conductor, heat is generated due to collision of free electrons with atoms.
- Let I amps be the strength of current for a potential difference of V volts across the conductor,

- Then the power absorbed by the resistor is given by

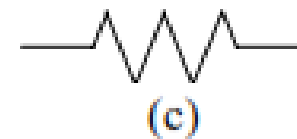
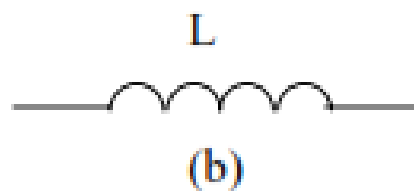
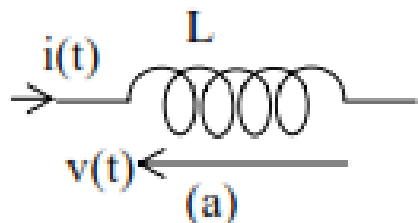
$$P = VI = (IR)I = I^2 R \text{ Watts}$$

- And the energy lost in the resistance in form of heat is then expressed as

$$w = \int_0^t P \cdot dt = Pt = I^2 Rt = \frac{V^2}{R} t$$

2 Inductance

- The inductance is the property of a material by virtue of which it opposes any change of magnitude or direction of electric current passing through the conductor.
- The unit of inductance being “henry” (L),
- It is given by Faraday’s law of electromagnetic induction.
- Inductance is said to be one henry when current through a coil of conductor changes at the rate of one ampere per second inducing one volt across the coil.



Cntd'

- A wire of finite length, when twisted into a coil, it becomes a simplest inductor.
- As soon as current will flow through the coil an electromagnetic field is formed.
- However, with any change of flow or direction of current, an electromagnetic field changes.
- This change of field induces the voltage (v) across the coil and is given by
- $v = L \frac{di}{dt}$
- Where i is the current through the inductor in ampere.

- The power absorbed by the inductor is given by

$$P = VI = Li \frac{di}{dt} \text{ Watts.}$$

- Energy absorbed by the inductor will thus be given by

$$W = \int_0^t P dt = \int_0^t Li \frac{di}{dt} dt = \frac{1}{2} Li^2$$

- From these equations, it is evident that the inductor can store finite amount of energy even the voltage across it may be null.

A pure inductor does not dissipate energy but only stores it.

3 Capacitance

- This is the capability of an element to store electric charge within it.
- A capacitor stores electric energy in the form of electric field being established by the two polarities of charges on the two electrodes of a capacitor.
- Quantitatively capacitance is a measure of charge per unit voltage that can be stored in an element.
- The unit of capacitance (C) is farad (F).
- The capacity is said to be one farad provided one coulomb of charge can be stored with one volt across the two electrodes of the element.



Capacitance calculation

- q being the amount of charge that can be stored in a capacitor of capacitance C against a potential difference of v volts, we can write
- $C = \frac{q}{v}$
- Then, $i = C \frac{dv}{dt}$ and $i = \frac{dq}{dt}$
- Where lower case letters represent the instantaneous values of respective parameters
- $dv = \frac{1}{C} i dt$

$$\text{or } \int_{v_o}^{v_f} dv = \frac{1}{C} \int_0^t i dt$$

Voltage current relationship of circuit elements

Circuit Element	Voltage (volts)	Current (amps)	Power (watts)
R (Ω)	$v = Ri$	$i = \frac{v}{R}$	$p = i^2 R$
L (H)	$v = L \frac{di}{dt}$	$i = \frac{1}{L} \int v dt + i_0$ [i_0 being the initial current]	$p = i \frac{di}{dt}$
C (F)	$v = \frac{1}{C} \int i dt + v_0$ [v_0 being the initial voltage]	$i = C \frac{dv}{dt}$	$p = Cv \frac{dv}{dt}$

Examples

EXAMPLE 1.1. A $100\ \Omega$ resistance is directly switched on across a 10 V battery. What is the current through the resistor? How much is the power loss? Also find the energy consumed in 5 secs.

SOLUTION. From the given data

$$V = 10\text{ V}, R = 100\ \Omega$$

$$\therefore I = \frac{V}{R} = \frac{10}{100} = 0.1\text{ A.}$$

$$\text{Power loss} = I^2 R = (0.1)^2 \times 100 = 1\text{ W}$$

$$\left[\text{Also Power loss} = \frac{V^2}{R} = \frac{10^2}{100} = 1\text{ W.} \right]$$

$$\begin{aligned} \text{Energy consumed} &= I^2 R t = 1 \times 5\text{ watts sec.} \\ &= 5\text{ joules.} \end{aligned}$$

EXAMPLE 1.1. Fig. E1.2 exhibits the voltage-time profile of a source to charge a capacitor of $50 \mu\text{F}$. What is the charging current?

SOLUTION. Here,

$$\frac{dv}{dt} = \frac{20}{10} = 2 \text{ V/m sec.}$$

$$= 2 \times 10^3 \text{ V/sec.}$$

$$\begin{aligned} \therefore i_{\text{charging}} &= C \frac{dv}{dt} = 50 \times 10^{-6} \times 2 \times 10^3 \\ &= 0.1 \text{ A.} \end{aligned}$$

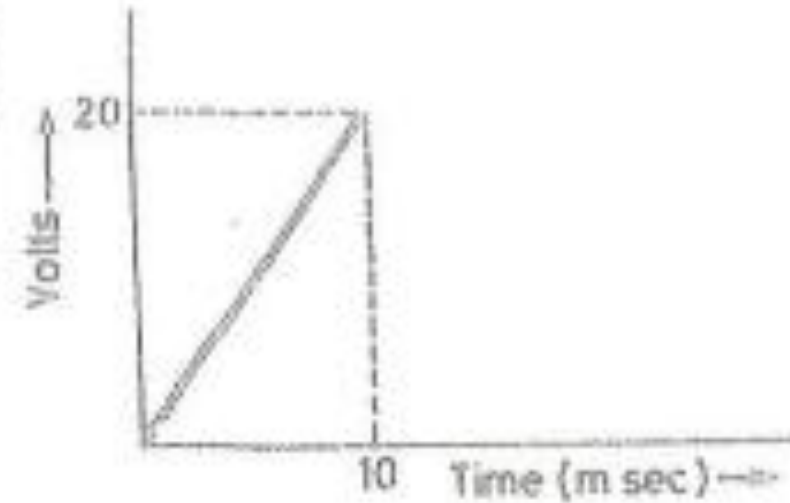
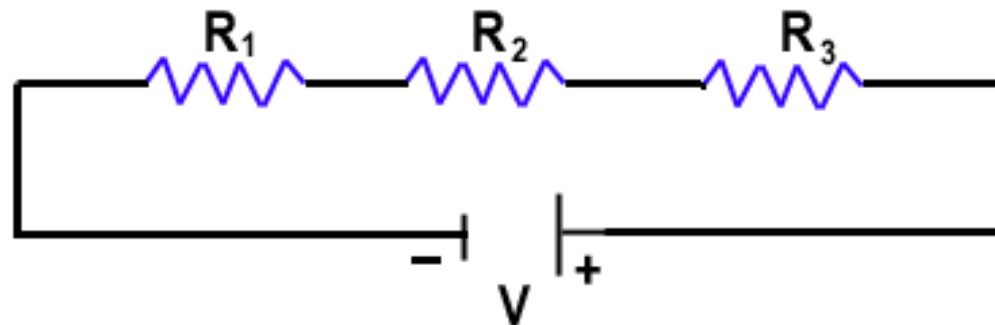


Fig. E1.2.

4. Series combination of Resistances

- Consider three resistances R_1 , R_2 , & R_3 connected in series combination with a power supply of voltage.
- The Potential difference of each resistor is V_1 , V_2 , & V_3 respectively.
- Let electric current I is passing through the circuit shown below.



$$R_e = R_1 + R_2 + R_3 + R_4 + \dots R_n$$

- Now

$$V = V_1 + V_2 + V_3$$

According to Ohm's law $V = IR$

- Then $IR = IR_1 + IR_2 + IR_3$

$$IR = I(R_1 + R_2 + R_3)$$

$$IR/I = R_1 + R_2 + R_3$$

$$R = R_1 + R_2 + R_3$$

- This shows that in series combination equivalent resistance of circuit is always greater than individual resistances.

$$\text{Since } I = \frac{V}{R}, \text{ then } I = \frac{V}{R_1 + R_2 + R_3}$$

- And the total power loss is given by

$$\begin{aligned} P &= I^2 R_1 + I^2 R_2 + I^2 R_3 \\ &= I^2 (R_1 + R_2 + R_3) \end{aligned}$$

Any voltage source connected to the terminals of a network of series resistors will provide the same current as if a single resistance, having a value of R_T , were connected between the open terminals. From Ohm's law we get

$$I = \frac{E}{R_T} \quad [\text{amps, A}]$$

The power dissipated by each resistor is determined as

$$P_1 = V_1 I = \frac{V_1^2}{R_1} = I^2 R_1 \quad [\text{watts, W}]$$

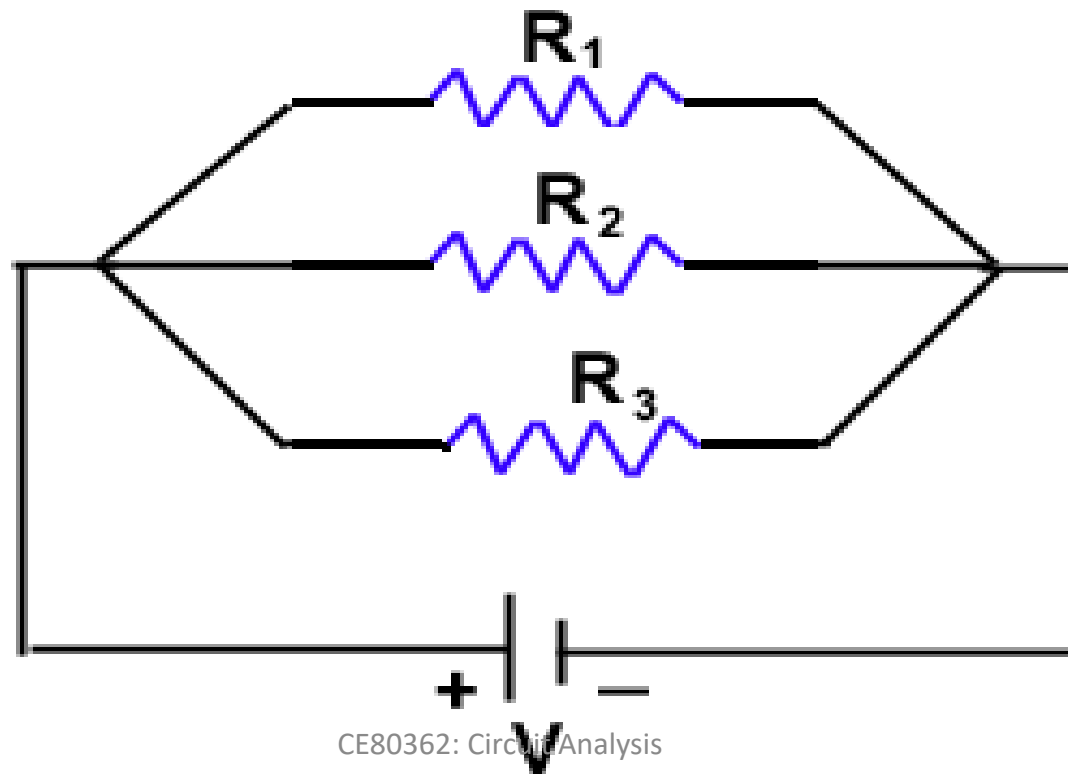
$$P_2 = V_2 I = \frac{V_2^2}{R_2} = I^2 R_2 \quad [\text{watts, W}]$$

$$\vdots$$

$$P_n = V_n I = \frac{V_n^2}{R_n} = I^2 R_n \quad [\text{watts, W}]$$

Parallel combination of Resistances

- Consider three resistances R_1 , R_2 & R_3 connected in parallel combination with a power supply of voltage V as shown below.



- Now

$$I = I_1 + I_2 + I_3$$

According to Ohm's law

$$V/R = I$$

Therefore,

$$V/R = V/R_1 + V/R_2 + V/R_3$$

$V/R = V (1/R_1 + 1/R_2 + 1/R_3)$ dividing by V we obtain

$$1/R = 1/R_1 + 1/R_2 + 1/R_3$$

An important effect of combining parallel resistors is that the resultant resistance will always be smaller than the smallest resistor in the combination.

Similarly, for the calculation of two resistances R_1 and R_2 connected in parallel:

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{\text{total}} = R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Calculation of the necessary parallel resistor R_2 , when R_1 and the total resistance R_{total} is given:

$$R_2 = \frac{R_1 \cdot R_{\text{total}}}{R_1 - R_{\text{total}}}$$

The power loss in the circuit above is given by

$$P = P_{R1} + P_{R2} + P_{R3}$$

P_{R1} , P_{R2} and P_{R3} being the individual power loss in the resistors R_1 , R_2 , R_3 respectively

$$\begin{aligned} P &= I_{R1}^2 \cdot R_1 + I_{R2}^2 \cdot R_2 + I_{R3}^2 \cdot R_3 \\ &= \left(\frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3} \right) \\ &= V^2 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \text{ Watts} \end{aligned}$$

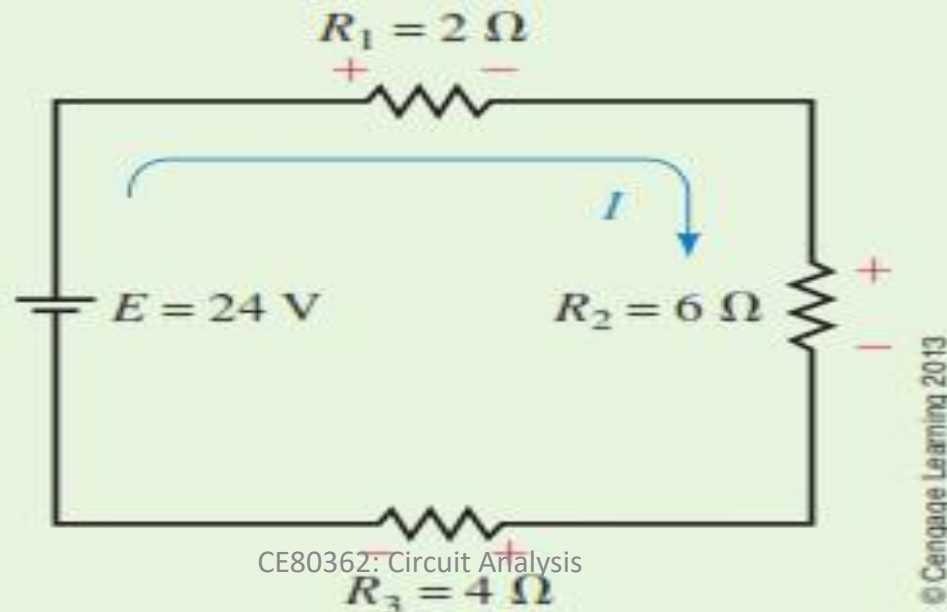
NOTE: The conductance G is defined as the reciprocal of resistance.

Thus: $G_T = G_1 + G_2 + G_3 + \dots + G_n$ equivalent to $1/R_T = 1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n$ and it is measured in siemens (S)

Example on series combination

For the series circuit shown in Figure 5–13, find the following quantities:

- Total resistance, R_T .
- Circuit current, I .
- Voltage across each resistor.
- Power dissipated by each resistor.
- Power delivered to the circuit by the voltage source.
- Verify that the power dissipated by the resistors is equal to the power delivered to the circuit by the voltage source.



Solution

a. $R_T = 2\ \Omega + 6\ \Omega + 4\ \Omega = 12.0\ \Omega$

b. $I = (24\ \text{V})/(12\ \Omega) = 2.00\ \text{A}$

c. $V_1 = (2\ \text{A})(2\ \Omega) = 4.00\ \text{V}$

$$V_2 = (2\ \text{A})(6\ \Omega) = 12.0\ \text{V}$$

$$V_3 = (2\ \text{A})(4\ \Omega) = 8.00\ \text{V}$$

d. $P_1 = (2\ \text{A})^2(2\ \Omega) = 8.00\ \text{W}$

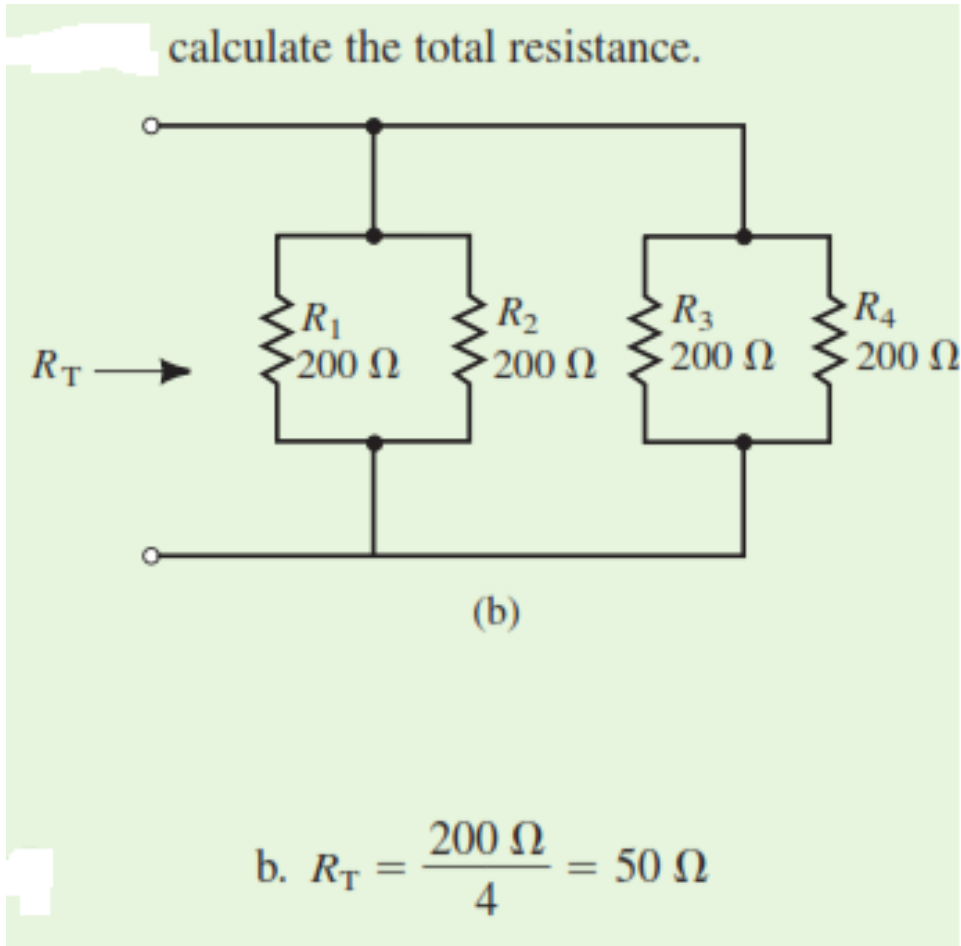
$$P_2 = (2\ \text{A})^2(6\ \Omega) = 24.0\ \text{W}$$

$$P_3 = (2\ \text{A})^2(4\ \Omega) = 16.0\ \text{W}$$

e. $P_T = (24\ \text{V})(2\ \text{A}) = 48.0\ \text{W}$

f. $P_T = 8\ \text{W} + 24\ \text{W} + 16\ \text{W} = 48.0\ \text{W}$

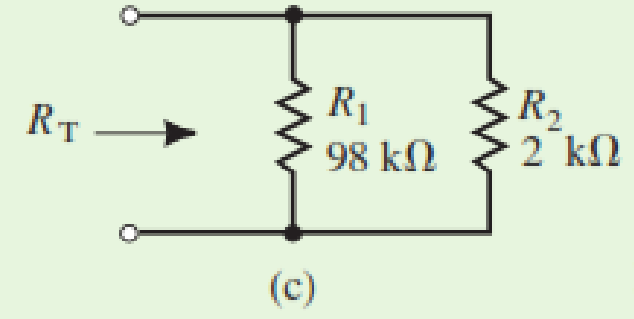
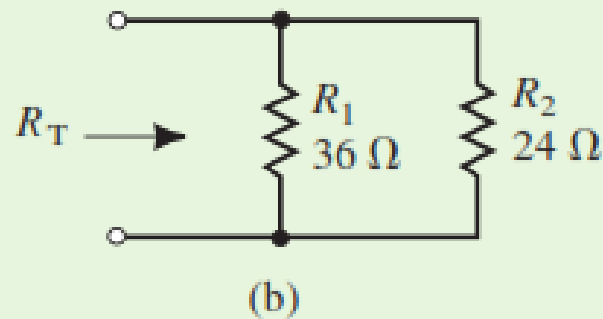
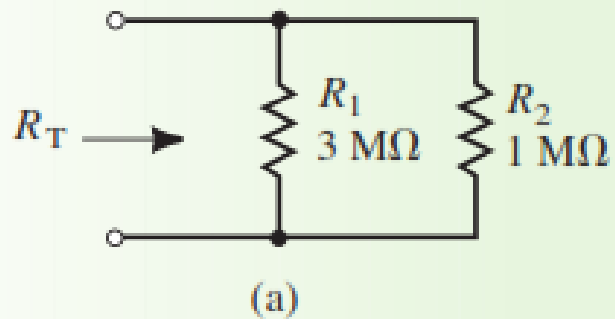
Example on parallel combination



- If the values of the resistors are equal. Just divide single value with the number of resistors as the lowest common multiple is one of the provided values.

Example on parallel combination for different values

Determine the total resistance of the resistor combinations of Figure 6–17.



Solution

$$\text{a. } R_T = \frac{(3 \text{ M}\Omega)(1 \text{ M}\Omega)}{3 \text{ M}\Omega + 1 \text{ M}\Omega} = 0.75 \text{ M}\Omega = 750 \text{ k}\Omega$$

$$\text{b. } R_T = \frac{(36 \text{ }\Omega)(24 \text{ }\Omega)}{36 \text{ }\Omega + 24 \text{ }\Omega} = 14.4 \text{ }\Omega$$

$$\text{c. } R_T = \frac{(98 \text{ k}\Omega)(2 \text{ k}\Omega)}{98 \text{ k}\Omega + 2 \text{ k}\Omega} = 1.96 \text{ k}\Omega$$

In this case, we follow the equation for calculation of resistors in parallel

Calculate the total resistance of the resistor combination of Figure 6–18.

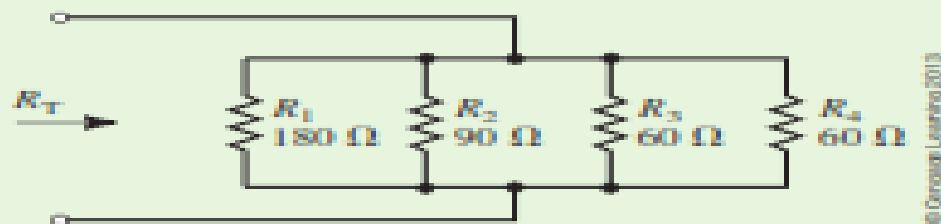


FIGURE 6–18

Solution By grouping the resistors into combinations of two, the circuit may be simplified as shown in Figure 6–19.

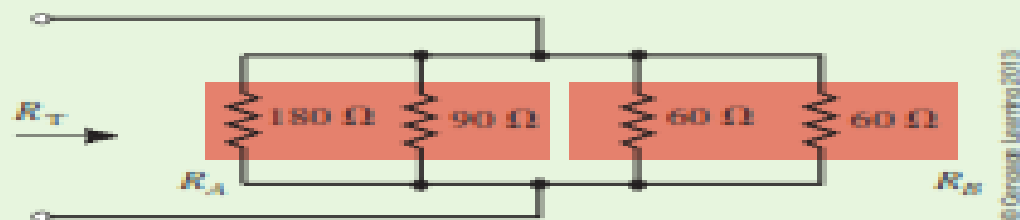


FIGURE 6–19

The equivalent resistance of each of the indicated combinations is determined as follows:

$$R_A = \frac{(180\ \Omega)(90\ \Omega)}{180\ \Omega + 90\ \Omega} = 60\ \Omega$$

$$R_B = \frac{(60\ \Omega)(60\ \Omega)}{60\ \Omega + 60\ \Omega} = 30\ \Omega$$

The circuit can be further simplified as a combination of two resistors shown in Figure 6–20.

The resultant equivalent resistance is

$$R_T = \frac{(60\ \Omega)(30\ \Omega)}{60\ \Omega + 30\ \Omega} = 20\ \Omega$$

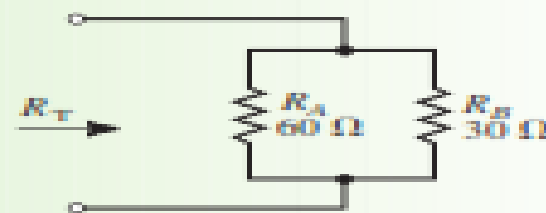


FIGURE 6–20

Note:

1. the resistors may be connected in both parallel and series in the same network.

In this case, series equation applies on series connection and parallel equation applies on parallel connection.

First make the circuit in series connection and then find the equivalent resistance.

Connection in Example 2 is different from example 3, so do not confuse them. First you need to understand how the source is connected to the entire circuit.

Series and Parallel combination of Capacitors

Capacitors in Series

For capacitors in series (Figure 10–21), the same charge appears on each. Thus, $Q = C_1V_1$, $Q = C_2V_2$, etc. Solving for voltages yields $V_1 = Q/C_1$, $V_2 = Q/C_2$, and so on. Applying KVL, we get $V = V_1 + V_2 + \dots + V_N$. Therefore,

$$V = \frac{Q}{C_1} + \frac{Q}{C_2} + \dots + \frac{Q}{C_N} = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}\right)$$

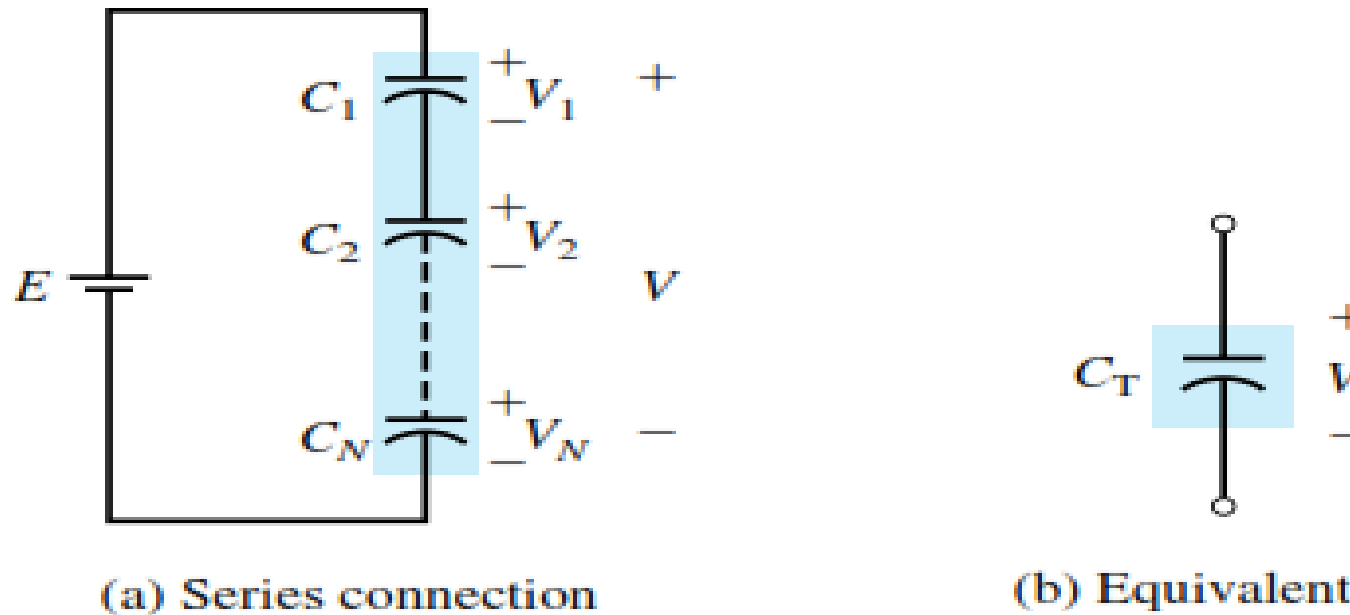


FIGURE 10–21 Capacitors in series: $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$.

But $V = Q/C_T$. Equating this with the right side and cancelling Q yields

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \quad (10-15)$$

For two capacitors in series, this reduces to

$$C_T = \frac{C_1 C_2}{C_1 + C_2} \quad (10-16)$$

For N equal capacitors in series, Equation 10-15 yields $C_T = C/N$.

EXAMPLE 10–7 Refer to Figure 10–22(a):

- Determine C_T .
- If 50 V is applied across the capacitors, determine Q .
- Determine the voltage on each capacitor.

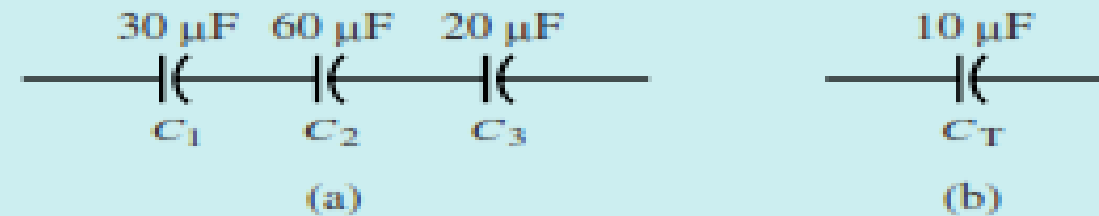


FIGURE 10–22

Solution

$$\begin{aligned}\text{a. } \frac{1}{C_T} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{30\ \mu\text{F}} + \frac{1}{60\ \mu\text{F}} + \frac{1}{20\ \mu\text{F}} \\ &= 0.0333 \times 10^6 + 0.0167 \times 10^6 + 0.05 \times 10^6 = 0.1 \times 10^6\end{aligned}$$

Therefore as indicated in (b),

$$C_T = \frac{1}{0.1 \times 10^6} = 10\ \mu\text{F}$$

$$\text{b. } Q = C_T V = (10 \times 10^{-6}\ \text{F})(50\ \text{V}) = 0.5\ \text{mC}$$

$$\text{c. } V_1 = Q/C_1 = (0.5 \times 10^{-3}\ \text{C})/(30 \times 10^{-6}\ \text{F}) = 16.7\ \text{V}$$

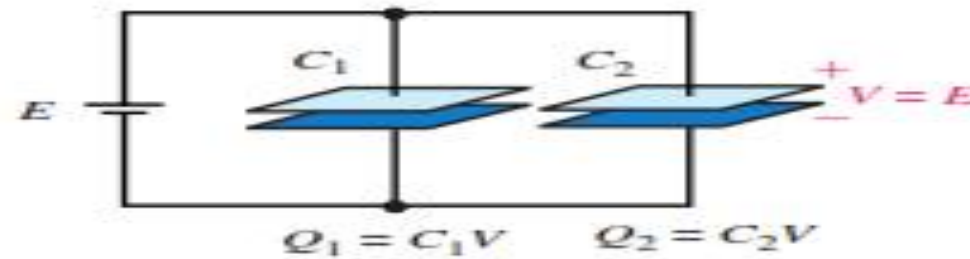
$$V_2 = Q/C_2 = (0.5 \times 10^{-3}\ \text{C})/(60 \times 10^{-6}\ \text{F}) = 8.3\ \text{V}$$

$$V_3 = Q/C_3 = (0.5 \times 10^{-3}\ \text{C})/(20 \times 10^{-6}\ \text{F}) = 25.0\ \text{V}$$

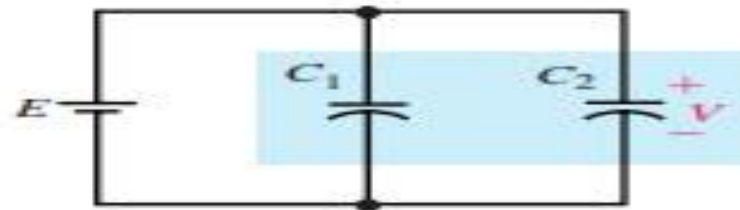
Check: $V_1 + V_2 + V_3 = 16.7 + 8.3 + 25 = 50\ \text{V}$.

Capacitors in Parallel

For capacitors in parallel, the effective plate area is the sum of the individual plate areas; thus, the total capacitance is the sum of the individual capacitances. This is easily shown. Consider Figure 10–20. The charge on each capacitor is given by Equation 10–1. Thus, $Q_1 = C_1V$ and $Q_2 = C_2V$. Since



(a) Parallel capacitors



(b) $C_T = C_1 + C_2$



$Q_T = Q_1 + Q_2$, $Q_T = C_1V + C_2V = (C_1 + C_2)V$. But $Q_T = C_TV$. Thus, $C_T = C_1 + C_2$. For more than two capacitors,

$$C_T = C_1 + C_2 + \cdots + C_N \quad (10-14)$$

That is, *the total capacitance of capacitors in parallel is the sum of their individual capacitances.*

EXAMPLE 10-6 A 10- μF , a 15- μF , and a 100- μF capacitor are connected in parallel across a 50-V source. Determine the following:

- Total capacitance.
- Total charge stored.
- Charge on each capacitor.

Solution

a. $C_T = C_1 + C_2 + C_3 = 10\ \mu\text{F} + 15\ \mu\text{F} + 100\ \mu\text{F} = 125\ \mu\text{F}$

b. $Q_T = C_TV = (125\ \mu\text{F})(50\ \text{V}) = 6.25\ \text{mC}$

c. $Q_1 = C_1V = (10\ \mu\text{F})(50\ \text{V}) = 0.5\ \text{mC}$

$$Q_2 = C_2V = (15\ \mu\text{F})(50\ \text{V}) = 0.75\ \text{mC}$$

$$Q_3 = C_3V = (100\ \mu\text{F})(50\ \text{V}) = 5.0\ \text{mC}$$

Check: $Q_T = Q_1 + Q_2 + Q_3 = (0.5 + 0.75 + 5.0)\ \text{mC} = 6.25\ \text{mC}$.

Series and Parallel combination of Inductances

For inductances in series or parallel, the equivalent inductance is found by using the same rules that you used for resistance. For the series case (Figure 13–15) the total inductance is the sum of the individual inductances:

$$L_T = L_1 + L_2 + L_3 + \cdots + L_N$$

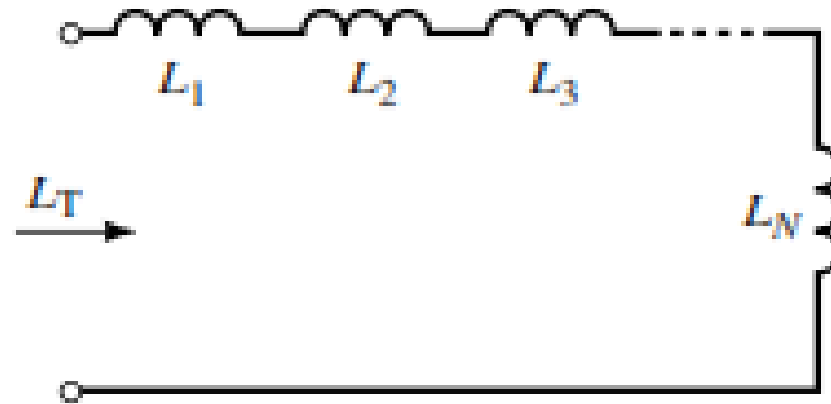


FIGURE 13–15 $L_T = L_1 + L_2 + \cdots + L_N$

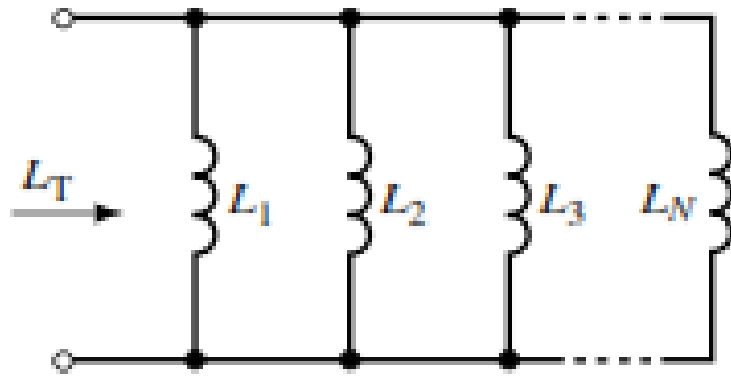


FIGURE 13–16 $\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$

For the parallel case (Figure 13–16),

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N}$$

For two inductances, Equation 13–9 reduces to

$$L_T = \frac{L_1 L_2}{L_1 + L_2}$$

examples

EXAMPLE 13–7 Find L_T for the circuit of Figure 13–17.

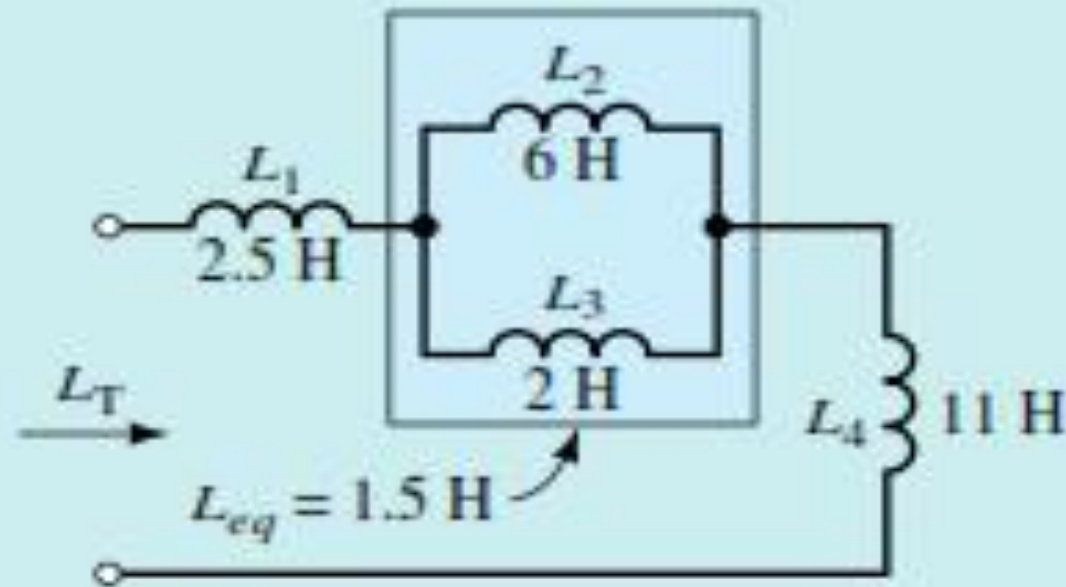


FIGURE 13–17

Solution The parallel combination of L_2 and L_3 is

$$L_{eq} = \frac{L_2 L_3}{L_2 + L_3} = \frac{6 \times 2}{6 + 2} = 1.5 \text{ H}$$

This is in series with L_1 and L_4 . Thus, $L_T = 2.5 + 1.5 + 11 = 15 \text{ H}$.

Reference books

1. Circuit Analysis. Theory & Practice, 5E by Wilhelm Miller
2. Circuit Analysis Theory and Practice 3rd ed
3. Circuit theory (Analysis and Synthesis), by A. Chakrabarti
4. A textbook of ELECTRICAL TECHNOLOGY, by B.L. Theraja and A.K. Theraja.
5. Other books, in library with almost the same contents.